

MICHAEL P. RAGSDALE

Systems Architect | Full-Stack Developer | Accessibility Specialist Norfolk, VA (Open to relocation within VA) | hireme@michaelpragsdale.com github.com/raggiesoft | michaelpragsdale.com

PROFESSIONAL SUMMARY

Systems Architect and Full-Stack Developer driven by a philosophy of "Empathetic Engineering." Specializing in building resilient, JSON-driven web architectures, custom Single Page Application (SPA) routers, and automated cloud media pipelines. Deeply committed to strict WCAG 2.1 AA and Section 508 compliance, creating performant technology that respects both system resources and the user's cognitive load.

TECHNICAL ARSENAL

- **Languages & Scripting:** C#, .NET 10, PHP 8.4, JavaScript (ES6+), Bash / Shell, Python, HTML5, CSS3/SCSS
- **Infrastructure & Cloud:** DigitalOcean (Droplets/Spaces), Nginx, Cloudflare (DNS/SSL), Certbot, Linux (Ubuntu/Debian)
- **Architecture & Automation:** Custom MVC Frameworks, Vanilla JS SPAs, FFmpeg Automation, Custom CI/CD pipelines
- **Compliance & Enterprise:** WCAG 2.1 AA, Section 508, Deque axe, ActiveNet CRM, Git, Microsoft 365

RELEVANT TECHNICAL EXPERIENCE

RaggieSoft Media | Norfolk, VA *Founder & Systems Architect* | Est. 2008 – Present

- **Frontend Architecture (SPA):** Architected and deployed a custom Vanilla JavaScript Single Page Application (SPA) router to replace standard browser navigation. Engineered seamless DOM-swapping, CSS variable synchronization, and History API state management to ensure gapless, uninterrupted HTML5 audio playback across a multi-domain web ecosystem.
- **Backend Frameworks (CMS):** Developed a proprietary, lightweight PHP 8.4 MVC content management system. Engineered a file-based, JSON-driven data architecture to eliminate database latency, serving complex narrative content and dynamic Schema.org (JSON-LD) structured data for advanced SEO indexing.
- **Cloud Infrastructure & Security:** Administer a cloud-based Linux environment (DigitalOcean Ubuntu) secured behind Cloudflare. Authored custom Nginx server block configurations to enforce a strict Single-Entry Point architecture, utilizing reverse proxies and hidden internal directories to mitigate automated scraping and privilege escalation.
- **CI/CD & Deployment Automation:** Designed a decoupled, Bash-based CI/CD pipeline featuring local synchronization agents and autonomous server-side deployment guardians. Enforced SetGID permissions and hash-based integrity checks against GitHub to execute zero-downtime atomic updates.
- **Media Automation (FFmpeg/Python):** Programmed automated FFmpeg shell scripts to recursively scan, calculate runtime, and transcode high-fidelity WAV masters into web-optimized tiered formats. Built Python utilities to ingest raw DOCX manuscripts and compile them into structured JSON for a WCAG-compliant web reader.
- **Systems Engineering (C#/.NET):** Engineered a proprietary .NET 10 ecosystem featuring a server-side integrity daemon and a MAUI desktop application, providing a strictly typed GUI for managing complex JSON data shards and eliminating manual syntax errors.

- **Data Sanitization & Accessibility:** Embedded strict WCAG 2.1 AA compliance into the framework layout, including ARIA live regions and dynamic focus management. Developed a data sanitization engine that preemptively prunes user-defined trigger terms from imported datasets to ensure a psychologically safe workflow.

Virginia Beach Parks & Recreation | Virginia Beach, VA *Operations Support & Technical Lead* | May 2023 – Present

- **Legacy Data Normalization:** Developed a manual ETL (Extract-Transform-Load) process bridging gaps in a legacy municipal reservation system. Converted raw scheduling data into Section 508 compliant, accessible operational documents, enriching output with cross-facility resource availability to optimize staff decision-making.
- **Technical Training & Onboarding:** Serve as the primary technical trainer for the ActiveNet CRM, instructing staff on data integrity protocols, payment processing security, and complex account management to ensure strict operational consistency.
- **Operational Leadership:** Function as a rotating Shift Supervisor, overseeing financial reconciliation for high-volume transactions. Act as the primary escalation point for resolving complex patron account issues and facility logistical conflicts.

ODU Online (Old Dominion University) | Norfolk, VA *Section 508 Compliance Specialist* | Nov 2019 – Jul 2022

- **Accessibility Compliance:** Ensured WCAG 2.1 AA and Section 508 compliance for online courses across enterprise LMS platforms, serving as a critical checkpoint before content publication.
- **Remediation & Testing:** Remediated complex web pages and documents, verifying student code and utilizing tools like Deque axe for detailed accessibility reporting.
- **Architectural Standardization:** Designed and developed a universal, accessible template for the University Policies section of the syllabus, significantly streamlining the workflow for instructional design faculty.

Residence Hall Association (ODU) | Norfolk, VA *Director of Administration* | Sep 2015 – May 2017

- **Digital Administration:** Administered the organization's Google Workspace environment, re-architecting a disorganized legacy file structure into a streamlined, secure hierarchy that optimized executive board workflows.

Dominion Enterprises | Norfolk, VA *Mobile Applications Developer Intern* | Sep 2012 – Dec 2012

- **Mobile Development:** Gained hands-on experience in cross-platform mobile development for Android & iOS using the Titanium SDK, successfully deploying a functional application to demonstrate architectural proficiency.

EDUCATION & CERTIFICATIONS

Tidewater Community College | Virginia Beach, VA *Associate of Science, Information Technology (Focus: Software Engineering & Web Dev)* | Expected Dec 2026

Old Dominion University | Norfolk, VA *Bachelor of Science, Customized Studies in Leadership (Focus: Project Management)* | In Progress

Wilson Workforce & Rehabilitation Center | Fishersville, VA *Certificate, Business & Information Technology* | 2011